



LECTURE 5

User Stories / Product Backlog

COEN / ELEC 390

Acknowledgement

- The concepts in these slides are based on Chapters 5 and 6 of the textbook. Please refer to these chapters for more information.

Essential Scrum, A Practical Guide to the Most Popular Agile Process
Kenneth S. Rubin, Addison-Wesley, 2013.

What Are User Stories?

- User stories are a simple way to describe what we want from a product, focusing on the value it will bring.
- They are designed to be clear to both business and technical teams.
- This simple approach aligns well with agile principles and helps us efficiently manage our product backlog.
- We can categorize user stories into 3 Cs: card, conversation, & confirmation.

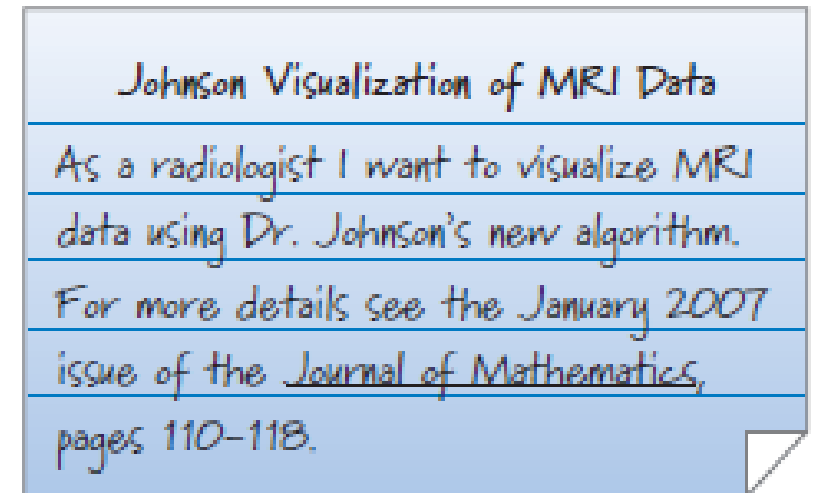
A. Card

- A card specifies a category of users (the user role), what that class of users wants to achieve (the goal), and why the users want to achieve the goal (the benefit)
- It serves as a placeholder for a more detailed discussion

User Story Title	Find Reviews Near Address
As a <user role> I want to <goal> so	As a typical user I want to see unbiased
that <benefit>.	reviews of a restaurant near an address
	so that I can decide where to go for
	dinner.

B. Conversation

- Conversations capture the details of the requirement.
- They are exposed and communicated in a conversation among the DT, PO and stakeholders
- The conversation is typically not a one-time event, but rather an ongoing dialogue.
- Conversations may lead to a UI sketch, further reading, etc.



Source: Essential Scrum, A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin, Addison-Wesley, 2013.

C. Confirmation

- A user story contains confirmation information in the form of conditions of satisfaction
 - The back of the card could specify the conditions of satisfaction
- Acceptance tests are an important way to capture and communicate, from the PO's perspective
 - They are a helpful way to create initial stories & refine them as more details become known
- Example: User story: Upload file; Conditions of satisfaction: Verify type of file (pdf, doc), maximum size, etc.

Type of User Stories

- Stories is the generic term
- Epics (very large stories)
 - Work spans months, maybe several releases
 - Describes a need at a high level of abstraction
 - A collection of features
- Features/theme (large stories)
 - Work spans more than 1 sprint
 - More detailed description of user need. May be part of an epic
 - A collection of stories or sprintable stories or implementable stories

Stories

- Stories or sprintable stories, implementable stories
 - Works spans less than one sprint
 - When these are put into a sprint they are broken into tasks (with estimated effort time)
- Tasks are typically done by one person
- Stories describe what to build. Tasks describe how to build it.
- Use Card, Conversation, Confirmation as the three “documents” for stories of any size (but level of details changes as the story size changes)

INVEST in Good Stories

Independent: User stories should be independent or at least only loosely coupled with one another.

Estimatable: Estimates provide an indication of the size and therefore the effort and cost of the stories.

Negotiable: Stories should encourage conversations where the details are negotiated among stakeholders.

Sized Appropriately (small): Stories worked on in sprints should be *small*. Weeks-long sprints >> days-long stories.

Valuable: Stories need to be valuable to a customer, user, or both.

Testable: Stories should be *testable* in a binary way—they either pass or fail their associated tests; means having good acceptance criteria.

Non-functional Requirements

- Non-functional requirements represent system-level constraints.
 - Example: performance, usability, reliability, testability, etc.
- They can be included in the user stories or the definition of done.
- If a non-functional requirements is included in a story then this will be met in a certain sprint. If in the definition of done then it is met in each sprint.

Gathering Stories

- Traditional approaches to requirements gathering involve asking the users what they want.
- A better approach is to involve the users as part of the team that is determining what to build and is constantly reviewing what is being built.
- This involves techniques such as:
 - User-story-writing workshops
 - Story Mapping

Story Mapping

- This approach focuses on the user to create detailed user stories.
- It starts by breaking down broad user activities into workflows, which are then divided into more specific tasks.
- This method provides a two-dimensional view of the Product Backlog.

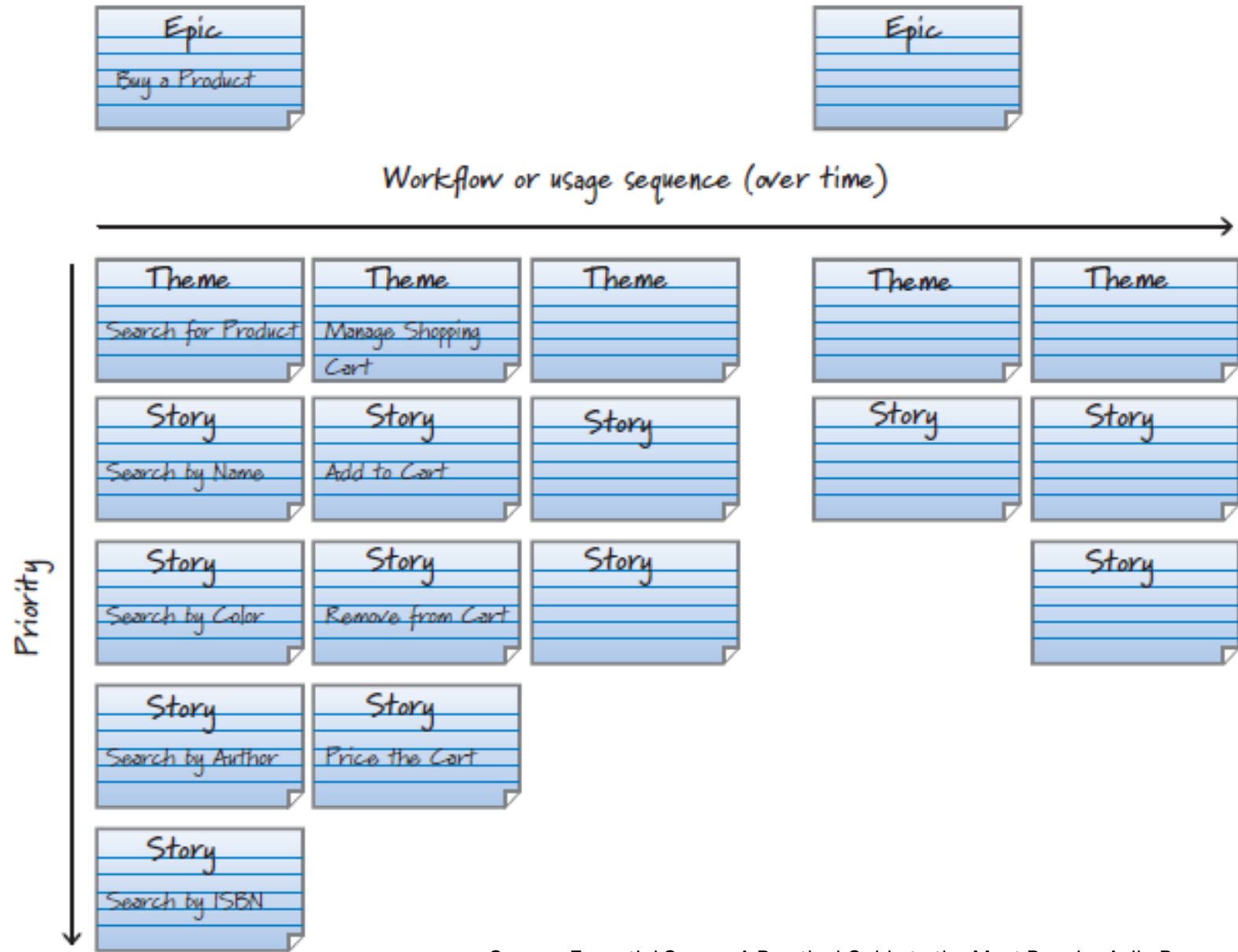


FIGURE 5.13 Story map



Product Backlog

COEN / ELEC 390 – Fall 2025

Product Backlog Items

- A product backlog item (PBI) represents a feature of the system
- Examples of PBIs include:
 - Adding a new feature
 - Changing to an existing feature
 - Fixing a bug
 - Improving the code
- For each product, there is exactly one product log

Good PB Characteristics (**D**EEP Principles)

- **Detailed Appropriately**

- PBIs that we plan to address soon should be positioned near the top of the backlog.
- They should be relatively small and highly detailed, making them suitable for inclusion in an upcoming sprint.
- As we approach the time to tackle larger PBIs, we will decompose them into smaller, sprint-ready tasks.
- It is essential to strike the right balance between having just enough detail and delivering stories just in time.

DEEP (Cont.)

- Emergent

- It is continuously updated based on a stream of economically valuable information.
- The PO must rebalance and reprioritize the product backlog, taking the new information into account.

- Estimated

- Each PBI has a size estimate corresponding to the effort required to develop it.

- Prioritized

- It is useful to prioritize the items that are destined for the next few sprints.

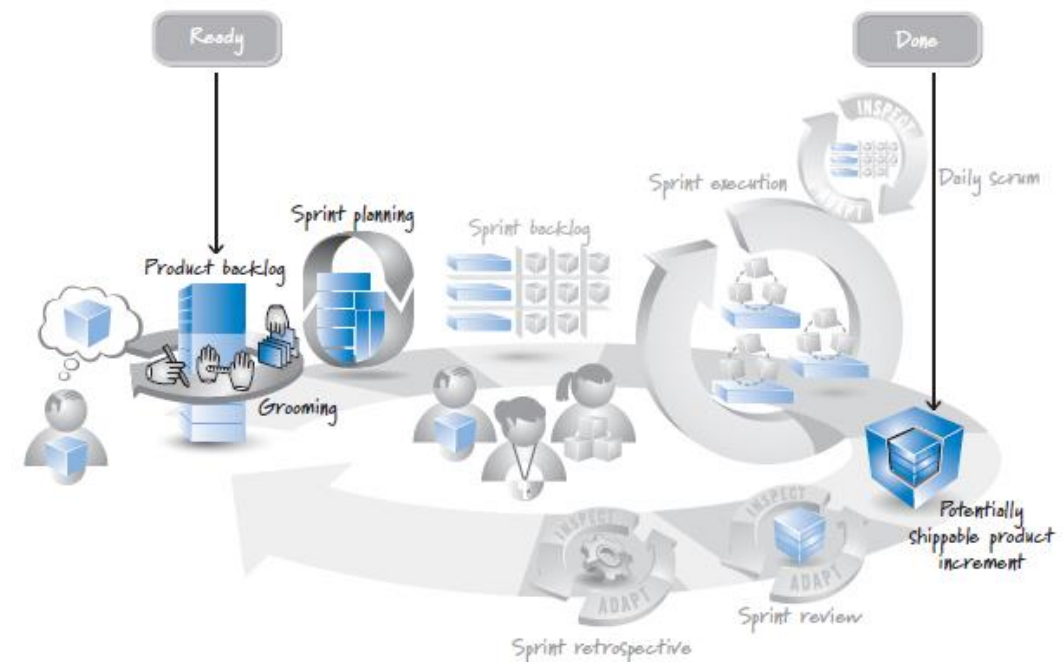
What is Grooming?

- Grooming is The process of creating and refining PBIs, estimating PBIs, and prioritizing PBIs
- Everyone is involved, lead by the PO
- Grooming can happen throughout the entire process

Definition of Ready

TABLE 6.2 Example Definition-of-Ready Checklist

Definition of Ready	
☐	Business value is clearly articulated.
☐	Details are sufficiently understood by the development team so it can make an informed decision as to whether it can complete the PBI.
☐	Dependencies are identified and no external dependencies would block the PBI from being completed.
☐	Team is staffed appropriately to complete the PBI.
☐	The PBI is estimated and small enough to comfortably be completed in one sprint.
☐	Acceptance criteria are clear and testable.
☐	Performance criteria, if any, are defined and testable.
☐	Scrum team understands how to demonstrate the PBI at the sprint review.



Source: Essential Scrum, A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin, Addison-Wesley, 2013.

Flow Management

- For a particular release, consider breaking PB into:
 - **Must-have features** are the essential items that are required for the upcoming release.
 - **Nice-to-have features** are those we aim to include in the next release; if we face time or resource constraints, some or all of these features may be dropped.
 - **Won't-have features** are those that will not be included in the current release.